Ramanujan Challenge 2026, Problem 3.1: the 7_2 regulator integral is $4\pi^2/85$

A proposed computer-assisted proof and independent audit

Robert Sneiderman
robbysneiderman@gmail.com

Abstract

Problem 3.1 of the Ramanujan Challenge asks for the exact value of a Godbillon–Vey type knot invariant: an integral of the Chern–Simons variation form along a distinguished real arc of the $\mathrm{SL}(2,\mathbb{C})$ character variety of the prime knot 7_2. We present a proposed computer-assisted proof that

$$\int_\alpha^\beta \left(\log x \,\frac{dy}{y} - \log y \,\frac{dx}{x}\right) = \frac{4\pi^2}{85}.$$

The strategy is to realize the whole arc, not just its endpoints, as an exact algebraic family of $\mathrm{SL}(2,\mathbb{C})$ representations, and then to recognize the integral as a difference of two *regulator* values — Cheeger–Chern–Simons invariants of the two closed manifolds obtained by Dehn filling at the ends of the arc. Concretely: we build an exact continuous real Riley and enhanced-Ptolemy lift of the full challenge branch; its left endpoint factors through the $(-1,2)$ Dehn filling of the 7_2 exterior and its right endpoint through the $(-1,1)$ filling. Four exact algebraic tetrahedron shapes at each endpoint give Neumann's extended Rogers sums S_α, S_β. A reciprocal Galois symmetry of the endpoint number fields, matched to an orientation-preserving symmetry of the four-tetrahedron triangulation, forces twice each extended Bloch class into a *totally real* number field; by Borel's rank theorem the relevant K-group is then finite, so each regulator value lies on an explicit finite lattice. A 1000-bit Arb interval evaluation on independently reconstructed algebraic shapes, with every logarithmic edge and Dehn-filling equation checked, is much narrower than one lattice spacing and therefore selects the exact values

$$S_\alpha = -\frac{26\pi^2}{15}, \qquad S_\beta = -\frac{86\pi^2}{51} \qquad (\mathrm{mod}\ \pi^2/2).$$

Zickert's Dehn-invariant formula and the differential of the extended Rogers dilogarithm identify the difference $S_\beta - S_\alpha$ with the required integral modulo $\pi^2/2$, and a coarse positive bound $0 < I < \pi^2/2$ removes the ambiguity in the lift. Direct quadrature and a separate signed-coordinate evaluation reproduce the result to about 60 and 180 decimal places, respectively, but neither numerical agreement is used as proof.

Nature of this proof

This is a proposed *computer-assisted* proof, and we want to be explicit about what that means here. The closing identity $\int_\alpha^\beta \eta = 4\pi^2/85$ (here $\eta = \log M \, d\log L - \log L \, d\log M$ is the Chern–Simons variation form defined in eq. (2) below) is verified numerically to about 180 decimal places; the logical proof does not rest on that numerical agreement but uses it only as a cross-check. The proof proper reduces the integral to a congruence on a finite, explicitly bounded lattice and then pins down the single lattice point by a certified interval computation. We have since independently re-derived the load-bearing pieces of this argument outside the original certificate's environment. The whole-arc lift, the single rational formula $u(M, L)$ of Section 2 that realizes the entire branch as one Riley family, was re-derived from scratch in plain `sympy`; the relator identity of Section 2, the enhanced-Ptolemy solution of Section 3, and the Dehn wedge $\nu = 2\,M \wedge L$ of Section 6 were each reproduced, symbolically or to high precision, from the printed equations alone; and the challenge integral itself was confirmed to sixty digits by a direct quadrature that touches none of this machinery. The two Chern–Simons regulator values S_α, S_β were then reproduced by an independent dilogarithm evaluator to about 150 digits, fed only the flattenings and gluing matrix read from SnapPy's public triangulation interface, so no step rests on the certificate's own Chern–Simons routine. Those two topological inputs have since been reconstructed by hand from the knot's four-tetrahedron triangulation, its explicit face pairings, using no private extended-Ptolemy solver: the shapes, flattenings, and gluing matrix are recovered from the gluing equations directly and reproduce the regulator values to over 160 digits. A separate 1000-bit Arb computation encloses the normalized regulator difference in a ball of diameter about 5×10^{-300} and selects $4/85$ as the unique point of the proved regulator lattice. The only residual appeal to SnapPy is to read public census combinatorics, the face pairings and the peripheral gluing rows, which the challenge permits.

Khoi's 2008 paper supplies the general Godbillon–Vey/A-polynomial strategy and evaluates related paths for the figure-eight and 5_2 knots, but it does not contain this 7_2 evaluation. The formula was presented as an open conjecture in the 2026 challenge. This manuscript was submitted to the organizers on July 12, 2026; organizer review remains pending. We therefore make no claim of acceptance, priority, or being a "first proof."

1 Notation and setup

This section fixes every symbol used later and gives, for each, a word about why it appears. A reader outside low-dimensional topology should be able to read the rest of the paper after this section. Two pieces of standard machinery — the extended Bloch group and the extended Rogers dilogarithm — are recalled where they are first needed, in Sections 4 and 6.

The curve and the two eigenvalues. The knot 7_2 has a well-defined $\mathrm{SL}(2,\mathbb{C})$ *character variety* $X = \mathrm{Hom}(\pi_1(S^3 \setminus 7_2), \mathrm{SL}(2,\mathbb{C}))//\mathrm{SL}(2,\mathbb{C})$, the space of representations of the knot group up to conjugacy, whose coordinate ring is generated by trace functions. Restricting a representation to the boundary torus and recording the meridian and longitude *eigenvalues* (M, L) (not their traces) sends the one-dimensional part of X to a plane curve in the (M, L)-plane; the defining polynomial of the Zariski closure of that curve is the knot's *A-polynomial* $A_{7_2}(M, L)$. The problem statement gives this polynomial explicitly as a degree-5 polynomial in L with coefficients in $\mathbb{Z}[M]$:

$$\begin{aligned} A_{7_2}(M,L) = L^5 &+ L^4\big(M^{14} - M^{12} + 3M^4 + 4M^2 - 2\big) \\ &+ L^3\big(-2M^{18} + 5M^{16} + M^{14} - 4M^{12} + 6M^8 + 5M^6 + 2M^4 - 4M^2 + 1\big) \\ &+ L^2\big(M^{22} - 4M^{20} + 2M^{18} + 5M^{16} + 6M^{14} - 4M^{10} + M^8 + 5M^6 - 2M^4\big) \\ &+ L\big(-2M^{22} + 4M^{20} + 3M^{18} - M^{10} + M^8\big) + M^{22}. \end{aligned} \tag{1}$$

Here M and L are the two coordinates on the curve, and they have a concrete meaning for a representation $\rho\colon \pi_1(S^3 \setminus 7_2) \to \mathrm{SL}(2,\mathbb{C})$. The boundary of the knot exterior is a torus with two distinguished loops, the *meridian* and the *longitude*. Their holonomies (the image matrices ρ(meridian) and ρ(longitude) under ρ) can be simultaneously upper-triangularized. This is not automatic for an arbitrary pair of matrices, but here the meridian and longitude are the two generators of the fundamental group $\mathbb{Z}^2$ of the boundary torus, so they commute in π_1; their holonomies therefore commute, and a commuting pair of matrices over $\mathbb{C}$ shares an eigenvector and can be conjugated into upper-triangular form simultaneously. (The two Riley meridian generators a, b of Section 2, by contrast, do *not* share an eigenvector: that failure is exactly the irreducibility of ρ.) In this triangular form,

- M is the *meridian-eigenvalue coordinate*: the function assigning to each representation ρ in the family an eigenvalue $M(\rho)$ of the meridian holonomy. At a single fixed ρ this is a definite number, but as ρ ranges over the one-dimensional character variety M varies, and it is this variation that the integrand records. It plays the role of the x-coordinate, $M = x$.
- L is the *longitude eigenvalue*: the corresponding eigenvalue of the longitude holonomy, in the challenge (= SnapPy standard-longitude) normalization. It is the y-coordinate, $L = y$.

The vanishing $A_{7_2}(M, L) = 0$ is the condition that a pair (M, L) of eigenvalues actually comes from a representation. This is the defining property of the A-polynomial [1]: A_{7_2} is (the nonabelian factor of) the defining polynomial of the Zariski closure of the eigenvalue pairs (M, L) realized by the peripheral holonomy of an $\mathrm{SL}(2, \mathbb{C})$ representation. We do not use this abstractly, and in particular do not rely on a point of $\{A_{7_2} = 0\}$ automatically lifting to a representation: the direction we actually need, that $A_{7_2}(M, L) = 0$ forces a genuine representation on the challenge branch, is established constructively in Section 2, where the Riley relator identity (11) exhibits an explicit irreducible representation at every point of the branch.

The Chern–Simons variation form η. The integrand of the challenge is the 1-form

$$\eta = \log M \, d\log L - \log L \, d\log M = \log x \, \frac{dy}{y} - \log y \, \frac{dx}{x}. \tag{2}$$

This is the real A-polynomial Chern–Simons variation form studied by Khoi [3]; the geometric content of the whole paper is that its integral along the arc below is a difference of Chern–Simons invariants. We write x, y and M, L interchangeably as dictated by context: x, y for the curve/integrand, M, L for the representation-theoretic eigenvalues.

The challenge branch and its endpoints. Among the real points of the curve there is one distinguished arc.

Definition 1 (challenge branch)**.** The *challenge branch* is the real branch $L = y(M)$ of $A_{7_2}(M, L) = 0$ for $\alpha \le M \le \beta$ on which y is positive and decreasing with slope $y'(x) \le -2$. For each x in $[\alpha, \beta]$ the quintic in L has five real roots; the challenge branch is the second-largest of them. The endpoints are

$$\alpha \approx 0.349269, \qquad \beta \approx 0.406813,$$

where α is the real root of $A_{7_2}(\alpha, \sqrt{\alpha}) = 0$ closest to 0.349269 and β is the real root of $A_{7_2}(\beta, \beta) = 0$ closest to 0.406813. The branch runs from $(M, L) = (\alpha, \sqrt{\alpha})$ to (β, β).

To high precision (50 digits),

$$\alpha = 0.34926850479999612520833503127884956848473645319259,$$

$$\beta = 0.40681308133678976238401142887341660342177298944918,$$

$$\tfrac{4\pi^2}{85} = 0.46445197181596981735691722352358358283829173681133.$$

Endpoint fields, and the roots t and r. The endpoints are algebraic, and we will need the number fields they generate. Call these the *endpoint fields.* They are the number fields generated by a real root of the two minimal polynomials

$$f_\alpha(t) = t^{12} - 3t^{11} + 4t^{10} - 5t^9 + 6t^8 - 7t^7 + 7t^6 - 7t^5 + 6t^4 - 5t^3 + 4t^2 - 3t + 1, \tag{3}$$

$$\begin{aligned} f_\beta(r) = {} & r^{16} - 7r^{15} + 22r^{14} - 48r^{13} + 87r^{12} - 133r^{11} + 178r^{10} - 211r^9 + 223r^8 \\ & - 211r^7 + 178r^6 - 133r^5 + 87r^4 - 48r^3 + 22r^2 - 7r + 1. \end{aligned} \tag{4}$$

The designated real roots are, plainly,

$$t = \sqrt{\alpha} = 0.59098942867025644049\ldots, \qquad r = \beta = 0.40681308133678976238\ldots. \tag{5}$$

That is: t is exactly the square root of the left endpoint's M-value, and r is exactly the right endpoint's M-value. The reason $t = \sqrt{\alpha}$ (and not α itself) is the asymmetric normalization in Definition 1: the left endpoint is defined by $L = M^{1/2}$, so its longitude eigenvalue is $\sqrt{\alpha} = t$ and its meridian eigenvalue is $\alpha = t^2$. Exact substitution gives the endpoint eigenvalue pairs

$$(M_\alpha, L_\alpha) = (t^2, t), \qquad (M_\beta, L_\beta) = (r, r), \qquad A_{7_2}(M_j, L_j) = 0, \tag{6}$$

where throughout, the subscript $j \in \{\alpha, \beta\}$ ranges over the two endpoints and (M_j, L_j) denotes the corresponding meridian/longitude eigenvalue pair. We write f_α, f_β for the polynomials (3)–(4) and $E_\alpha = \mathbb{Q}(t)$, $E_\beta = \mathbb{Q}(r)$ for the endpoint fields they generate.

Riley representations. The knot 7_2 is the two-bridge knot $b(11, 5)$ (Schubert normal form). For a two-bridge knot, every irreducible $\mathrm{SL}(2, \mathbb{C})$ representation of the knot group is conjugate to a two-parameter *Riley representation*: fix one meridian in upper-triangular form and normalize the conjugate meridian, obtaining generators

$$a = \begin{pmatrix} s & 1 \\ 0 & s^{-1} \end{pmatrix}, \qquad b = \begin{pmatrix} s & 0 \\ -u & s^{-1} \end{pmatrix}, \tag{7}$$

where $s = M$ is the meridian eigenvalue and u is the *Riley parameter*, the single off-diagonal unknown chosen so that (a, b) satisfies the knot-group relation. Fixing the meridian eigenvalue $s = M$, that relation becomes the Riley polynomial $\phi(M, u) = 0$, which has finitely many roots u; each root gives a representation with its own longitude eigenvalue L, so along the A-polynomial curve $A_{7_2}(M, L) = 0$ exactly one value of u is selected at each point (M, L). Thus u is a single-valued rational function $u = u(M, L)$ of the peripheral eigenvalue pair, given explicitly in (10) below. (Warning: the letter b is reused in Section 3 for an unrelated Ptolemy variable.) The relation and the longitude are built from the relator word

$$w = aba^{-1}b^{-1}aba^{-1}b^{-1}ab \quad \text{(10 letters)}, \qquad \lambda = w^* w a^{-4}, \tag{8}$$

where w^* is w read backwards. The knot-group relation is $wa = bw$, and λ is the canonical longitude of this two-bridge presentation. The challenge longitude is the *inverse* of λ, so if λ_{11} denotes the $(1, 1)$ entry (an eigenvalue) of the longitude matrix, then the challenge longitude eigenvalue is

$$L = \lambda_{11}^{-1}, \qquad \text{equivalently } L\lambda_{11} = 1. \tag{9}$$

The identity $L\lambda_{11} = 1$ is the "basis bridge" between the two-bridge presentation and the SnapPy standard basis, and we use it again in Section 6.

Symbols introduced later. For orientation, the remaining notation is local to its section and defined there: the enhanced Ptolemy coordinates ($c_{1100,0}$, $m = M^{-1}$, and the variables b, c, e) and cross-ratios $z_{j,0}, \ldots, z_{j,3}$ in Section 3; the shape parameters $z_0, \ldots, z_3$, the reciprocal involutions τ, the fixed-field generators $X = t + t^{-1}$ resp. $r + r^{-1}$, the fixed fields F_j, the extended Bloch classes x_j, the triangulation automorphism ϕ, and the torsion order $w(F)$ in Section 4; the logarithmic coordinates w_0, w_1, w_2 and the regulator sums S_α, S_β in Section 5; and the extended Rogers function $R(z)$, the Dehn invariant ν, and the log-coordinates $X = -\log x$, $Y = -\log y$ in Section 6. A handful of letters are deliberately reused (the matrix I vs. the integral I; the Riley b vs. the Ptolemy b; the polynomial $R(s, u)$ vs. the Rogers function $R(z)$; X as a field generator vs. $X = -\log x$; the local exponent ν_p vs. the Dehn invariant ν); each reuse is flagged at its second appearance.

2 The curve and its two representations

The goal of this section is to promote the two *endpoints* of the challenge branch to genuine $\mathrm{SL}(2, \mathbb{C})$ representations, and in fact to lift the *entire* arc as one algebraic family. Recognizing the endpoints as representations of closed Dehn fillings is what later lets us treat the integral as a difference of Chern–Simons invariants. Everything here is done by exact rational-function arithmetic, independently of any triangulation.

One rational formula for the whole arc. The key point — and the step flagged in the honesty note — is that a single rational function of (M, L) gives the Riley parameter along the whole branch. Set $s = M$ and

$$u(M, L) = \frac{(M^2 - 1)(L - 1)}{M^2 + L}. \tag{10}$$

With this choice, direct rational-function arithmetic in the Riley matrices (7) evaluates the relator $wa - bw$ explicitly. Its $(1, 2)$ entry is

$$(wa - bw)_{12} = \frac{A_{7_2}(M, L)}{M^6(M^2 + L)^5}, \qquad A_{7_2}(M, L) \mid \mathrm{num}(L\lambda_{11} - 1) \quad \text{in } \mathbb{Q}[M, L], \tag{11}$$

and the other three relator entries either vanish identically or are multiples of this one. In particular, on the curve $A_{7_2}(M, L) = 0$ the relation $wa = bw$ holds, so (10) really does produce a representation for every point of the branch. The same arithmetic yields the commutator trace

$$\mathrm{tr}([a, b]) - 2 = \frac{(L - 1)(M^2 - 1)^2(L - M^4)}{M^2(M^2 + L)^2}, \tag{12}$$

where $[a, b] = aba^{-1}b^{-1}$. Since $\mathrm{tr}([a, b]) - 2 \neq 0$ exactly when the representation is irreducible, (12) is our irreducibility detector.

Proposition 1. *On the challenge branch one has $1/4 < M < 1/2$, $1/4 < L < 1$, and $L > M^4$. Hence every denominator in (10)–(12) is nonzero, and (12) never vanishes. Therefore (10) defines a continuous real irreducible representation path over the whole challenge arc, with positive peripheral eigenvalues different from 1, i.e. purely hyperbolic peripheral holonomy.*

Proof. The stated inequalities are exact facts about the branch (they follow from the root isolation recorded in Section 6). Given them, the denominators $M^2 + L$, M^2, and $M^6(M^2 + L)^5$ are all nonzero, so (10) and the relator identity (11) are valid; and $(L - 1)(M^2 - 1)^2(L - M^4) \neq 0$ because $L \neq 1$, $M \neq \pm 1$, and $L > M^4$, so (12) is nonzero and the representation is irreducible.

The Riley normalization (7) is itself elementary: triangularize one meridian, then irreducibility lets one normalize the conjugate meridian as shown. A reducible knot representation would have longitude eigenvalue 1, because its diagonal character factors through the abelianization, on which the longitude is trivial; here $L \neq 1$, consistent with irreducibility. □

Isolating the two endpoint characters and their fillings. Let $R(s,u)$ be the numerator of the $(1,2)$ entry of $wa - bw$. (This $R(s,u)$ is a polynomial; it is unrelated to Neumann's Rogers function $R(z)$ of Section 6.) The diagonal entries of $wa - bw$ vanish identically and its $(2,1)$ entry has numerator $uR(s,u)$. Running the exact Euclidean algorithm in the variable u over each endpoint field isolates a single value of u, via linear greatest common divisors:

$$\begin{aligned}
\gcd_u\big(R(t^2,u), t\lambda_{11}(t^2,u) - 1\big) &= u + 2t^{11} - 5t^{10} + 5t^9 - 7t^8 + 9t^7 - 9t^6 \\
&\quad + 9t^5 - 10t^4 + 7t^3 - 6t^2 + 5t - 2, \qquad (13)\\
\gcd_u\big(R(r,u), r\lambda_{11}(r,u) - 1\big) &= u + r^{15} - 7r^{14} + 22r^{13} - 48r^{12} + 87r^{11} \\
&\quad - 133r^{10} + 178r^9 - 211r^8 + 223r^7 - 211r^6 + 178r^5 \\
&\quad - 133r^4 + 87r^3 - 48r^2 + 21r - 5. \qquad (14)
\end{aligned}$$

Substituting the resulting root for u makes *every* entry of $wa - bw$ zero, and moreover gives the exact matrix identities

$$a^{-1}\lambda^{-2} = I \quad (\alpha), \qquad a^{-1}\lambda^{-1} = I \quad (\beta), \qquad (15)$$

where I is the 2×2 identity matrix. (This I is unrelated to the integral I of Section 6.) These are precisely the relations of the closed manifolds obtained by Dehn surgery. (Dehn filling glues a solid torus to the torus boundary of the exterior so that the peripheral curve $p\cdot$(meridian)$+q\cdot$(longitude) becomes contractible; the resulting closed manifold's group is the knot group with that curve killed, which is exactly a relation of the form (15).) The left representation factors through the SnapPy filling $(-1,2)$, and the right through $(-1,1)$. One checks directly that the two gcd roots (13)–(14) are the specializations of the single generic formula (10) at (t^2,t) and (r,r); so the endpoint characters are exactly the ends of the one-parameter family of Proposition 1. Finally, (9) records the basis bridge $L\lambda_{11} = 1$: the variable L is the challenge (and SnapPy standard-longitude) eigenvalue, while λ is the canonical longitude of the two-bridge presentation.

Remark 1. The whole-arc lift (10), together with the identities (11)–(12), was independently re-derived from scratch in a separate computer algebra system; see the note at the head of the paper.

3 Exact enhanced Ptolemy points

To compute Chern–Simons invariants we need the representations expressed on an actual ideal triangulation, as tetrahedron shapes. We use SnapPy's four-tetrahedron triangulation of the 7_2 exterior together with Zickert's *enhanced Ptolemy variety* [6]: an affine variety whose points parametrize generically decorated $\mathrm{SL}(2,\mathbb{C})$ representations on a *fixed* triangulation with prescribed meridian and longitude eigenvalues. Here a *decoration* (also called a boundary-decoration) is a choice of horosphere — equivalently a coset fixing a lift of the peripheral fixed point — at each cusp; this is the extra boundary data the Ptolemy coordinates record. Its coordinates are 24 signed numbers, one per oriented edge-with-sign; from them one reads off the tetrahedron shapes, and from the shapes the Cheeger–Chern–Simons regulator. We recover the same characters as in Section 2 on this triangulation.

The gauge-fixed system. Normalize one enhanced Ptolemy coordinate to $c_{1100,0} = 1$ (a gauge choice), abbreviate $m = M^{-1}$, and eliminate the coordinates identified by the face pairings. The enhanced ideal then reduces to a four-equation system in three unknowns b, c, e:

$$\begin{aligned} -Mb - mc - be &= 0, \\ -Lmb^2 - m - e &= 0, \\ Lm^4 b + Lmbe - c &= 0, \\ Mc^2 - Me^2 + c &= 0. \end{aligned} \tag{16}$$

Here b, c, e are Ptolemy coordinate unknowns and have *nothing* to do with the Riley generator b of (7). This system lifts the whole real arc on one fixed triangulation. On the curve $A_{7_2}(M, L) = 0$ the solution is

$$b^2 = \frac{(1 - M^2)(L - M^4)}{LM^2(M^2 + L)}, \qquad e = -\frac{Lb^2 + 1}{M}, \qquad c = b(Lb^2 + 1 - M^2). \tag{17}$$

The certificate uses an explicit rational choice of $b(M, L)$ and verifies all four equations of (16) as rational-function identities modulo A_{7_2}. As the sharpest such check, the difference between its b^2 and the displayed b^2 of (17) is exactly

$$\frac{(M^6 + L)\, A_{7_2}(M, L)}{L^2(M^2 - 1)^2 M^6 (M^2 + 1)^2 (M^2 + L)^4},$$

which vanishes on the curve.

Choosing a continuous branch of shapes. The displayed b^2 in (17) is positive on the challenge arc, and both endpoint solutions have $b < 0$, so we take the continuous negative branch. Then

$$Lb^2 + 1 - M^2 = \frac{L(1 - M^2)(1 + M^2)}{M^2(M^2 + L)} > 0,$$

so by (17) the coordinates b, c, e stay negative and nonzero along the arc. Each of the 24 signed coordinates is a fixed sign times a monomial in M, L times one of $1, b, c, e$; hence none vanishes, the four tetrahedron shapes avoid 0 and 1, and their logarithms can be continued from the left endpoint to the right without a branch change. This continuity is what makes the endpoint regulators comparable in Section 5.

Exact endpoint solutions and identification with the Riley characters. Over $E_\alpha = \mathbb{Q}(t)$ at $(M, L) = (t^2, t)$ and over $E_\beta = \mathbb{Q}(r)$ at $(M, L) = (r, r)$, the reduced Gröbner basis of (16) is linear in b, c, e. The certificate solves it exactly, expands all 24 signed Ptolemy coordinates, and substitutes them back into the unreduced enhanced Ptolemy ideal; every residual is the zero element of the relevant number field. Writing $z_{j,0}, \ldots, z_{j,3}$ for the four primary cross-ratios (tetrahedron shape parameters) obtained from those coordinates at endpoint j, direct multiplication in the number fields gives every edge product and the filled-cusp product equal to 1 exactly, and no shape equals 0 or 1. So the Ptolemy point is a nondegenerate decorated representation with the stated peripheral eigenvalues, and its three base coordinates match the specializations of the rational full-curve coordinates (17).

These endpoint Ptolemy representations are the endpoint Riley representations of Section 2. Indeed $L_j \neq 1$, whereas a reducible knot-group representation has longitude eigenvalue 1; so the

Ptolemy representation is irreducible, hence conjugate to the Riley form (7), and the linear gcds (13)–(14) leave exactly one character over each peripheral point. This proves equality of the two *characters*, not merely of their peripheral eigenvalues.

Remark 2. The original certificate obtained this system through SnapPy's enhanced Ptolemy interface. The public audit no longer treats that interface as an oracle: one checker verifies the printed equations, while a second reconstructs the edge equations, shapes, and flattenings from the explicit four-tetrahedron census triangulation. The remaining SnapPy inputs are public face-pairing and peripheral-curve data. Both checkers are included in the verification bundle.

4 Reciprocal symmetry and a finite regulator lattice

We now recall just enough K-theory to say what a "regulator" is here, and then exploit a symmetry to trap it on a finite lattice. Concretely, the regulator of an endpoint representation is the complex number S_j computed in Section 5 as a sum of extended Rogers dilogarithm values over the four tetrahedron shapes; it is well-defined only modulo the period recorded in the table below, and the whole proof consists of pinning down that number. With that picture fixed, the K-theory that follows is structure, not definition. Neumann's *extended Bloch group* $\widehat{\mathcal{B}}(F)$ of a field F is the receptacle for a Cheeger–Chern–Simons class of a representation together with a choice of flattening (a lift of tetrahedron shape logarithms); it is naturally isomorphic to the indecomposable part $K_3^{\mathrm{ind}}(F)$ of algebraic K-theory, and it carries the extended Rogers regulator (defined in Section 6). There is an SL version $\widehat{\mathcal{B}}_{\mathrm{SL}}(F)$ and a PSL version, differing by the period conventions recorded below. The value $\int \eta$ we want will be a difference of regulator values, so the whole difficulty is to know that difference lies in a small, explicitly known group.

Reciprocal involutions and totally real fixed fields. Both endpoint polynomials (3)–(4) are reciprocal, so $t \mapsto t^{-1}$ and $r \mapsto r^{-1}$ are field automorphisms. Let $\tau_\alpha(t) = t^{-1}$ on E_α and $\tau_\beta(r) = r^{-1}$ on E_β; write τ for either one. Its fixed field is generated by $X = t + t^{-1}$ (resp. $X = r + r^{-1}$), with defining polynomials

$$g_\alpha(X) = X^6 - 3X^5 - 2X^4 + 10X^3 - X^2 - 7X + 1, \tag{18}$$

$$g_\beta(X) = X^8 - 7X^7 + 14X^6 + X^5 - 25X^4 + 9X^3 + 12X^2 - 3X - 1. \tag{19}$$

(This field generator X is unrelated to $X = -\log x$ in Section 6.) Exact real-root isolation gives signatures $(6, 0)$ and $(8, 0)$ and discriminants $5^3 \cdot 3881$ and 17^7; in particular both fixed fields, which we call F_α and F_β, are *totally real*.

Matching the field symmetry to a symmetry of the triangulation. The reciprocal action on the exact tetrahedron shapes is the same at both endpoints. Writing z_0, z_1, z_2, z_3 for the four generic shape parameters (recall a shape z has partners $1 - z^{-1}$, etc.),

$$\tau(z_0) = 1 - z_2^{-1}, \qquad \tau(z_1) = z_1, \qquad \tau(z_2) = (1 - z_0)^{-1}, \qquad \tau(z_3) = z_3. \tag{20}$$

The four-tetrahedron triangulation has a nonidentity orientation-preserving automorphism ϕ,

$$0 \mapsto (2; (2,1,3,0)), \quad 1 \mapsto (1; (3,2,1,0)), \quad 2 \mapsto (0; (3,1,0,2)), \quad 3 \mapsto (3; (1,0,3,2)), \tag{21}$$

where the tuple after each semicolon is the vertex permutation. All four permutations are even (so ϕ preserves orientation), and the induced cusp map is $-I$, so ϕ extends over both fillings. Its shape transformations are exactly (20). We write ϕ_* for the induced map on representations.

Twice each Bloch class descends to the totally real field. The exact Riley lift is an $\mathrm{SL}_2(E_j)$ representation of the closed filling, so it defines an extended Bloch class

$$x_j \in \widehat{\mathcal{B}}_{\mathrm{SL}}(E_j) \simeq K_3^{\mathrm{ind}}(E_j),$$

with image $\bar{x}_j$ in the PSL extended Bloch group. The two groups and their periods must be kept separate:

group / class	period	reference
SL regulator	mod $4\pi^2$	[7, Section 2.2.1]
PSL Cheeger–Chern–Simons	mod π^2	[4, Prop. 2.5, Thm. 2.6]
certified evaluator	mod $\pi^2/2$	[5, Thm. 3.4]

All maps here are homomorphisms, so an order bound for x_j also bounds the evaluator's value on $\bar{x}_j$. Now the shape identities (20) say that $\tau_j\rho_j$ is PSL_2-conjugate to $\rho_j \circ \phi_*$, where ρ_j is the endpoint representation and ϕ is the automorphism (21). Nondegenerate shape data determine the decorated PSL representation; ϕ preserves orientation and its cusp map is $-I$, so it extends over each filling and preserves the closed fundamental class. The two resulting SL lifts can therefore differ only by a $\mathbb{Z}/2$ cocycle, and Zickert's two-torsion result for such a change of lift gives [7, Theorem 6.15]

$$2\tau_j(x_j) = 2x_j.$$

Galois descent [7, Corollaries 7.14–7.15] then places $2x_j$ in $K_3^{\mathrm{ind}}(F_j)$, where F_j is the totally real fixed field of τ_j.

Borel finiteness and the explicit lattice. Borel's rank theorem states rank $K_3^{\mathrm{ind}}(F) = r_2(F)$, the number of complex (conjugate-pair) places of F [2]. Since F_j is totally real, $r_2(F_j) = 0$, so $K_3^{\mathrm{ind}}(F_j)$ is *finite*. Its torsion subgroup is cyclic of order

$$w(F) = 2\prod_p p^{\nu_p}, \qquad \nu_p = \max\{\nu : \zeta_{p^\nu} + \zeta_{p^\nu}^{-1} \in F\} \tag{22}$$

[7, Section 1], where ζ_{p^ν} is a primitive p^ν-th root of unity and $\zeta + \zeta^{-1}$ is the real generator of its cyclotomic field. (This local exponent ν_p is unrelated to the Dehn invariant ν of Section 6.) We need only degree-based upper bounds: if F is totally real of degree $d = [F : \mathbb{Q}]$ and contains $\zeta_{p^\nu} + \zeta_{p^\nu}^{-1}$, then $\varphi(p^\nu)/2$ divides d (for $p^\nu > 2$), with φ Euler's totient. Applying this to F_α (degree 6): the condition $\varphi(p^\nu) \mid 12$ is met only by the prime powers $p^\nu = 8$ ($\varphi = 4$), 9 ($\varphi = 6$), 5 ($\varphi = 4$), 7 ($\varphi = 6$), 13 ($\varphi = 12$), giving factors $2^3, 3^2, 5, 7, 13$; for F_β (degree 8) the condition $\varphi(p^\nu) \mid 16$ is met only by 32 ($\varphi = 16$), 3 ($\varphi = 2$), 5 ($\varphi = 4$), 17 ($\varphi = 16$), giving $2^5, 3, 5, 17$. With the leading factor 2 of (22) this yields

$$w(F_\alpha) \mid 2 \cdot 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 13 = 65520, \qquad w(F_\beta) \mid 2 \cdot 2^5 \cdot 3 \cdot 5 \cdot 17 = 16320. \tag{23}$$

Thus x_α and x_β have orders dividing 131040 and 32640. After projection to $\mathbb{C}/(\pi^2/2)\mathbb{Z}$, the regulator difference has order dividing

$$N = \mathrm{lcm}(131040, 32640) = 8910720,$$

so, after division by π^2, it lies on the finite lattice

$$\frac{1}{2N}\mathbb{Z}\Big/\frac{1}{2}\mathbb{Z} = \frac{1}{17821440}\mathbb{Z}\Big/\frac{1}{2}\mathbb{Z}. \tag{24}$$

5 Certified selection of the regulator value

The regulator difference is now known to sit on the lattice (24), whose spacing is about 5.61×10^{-8}. It remains to identify the exact lattice point. We do this by a certified 1000-bit Arb interval evaluation of the extended Rogers sum — the sum of Neumann's extended Rogers function $R(z)$ (defined in Section 6) over the four tetrahedra — at each endpoint, narrow enough to pin one point.

Flattenings with no hidden $2\pi i$. For each endpoint, the independently reconstructed algebraic shapes determine logarithmic coordinates w_0, w_1, w_2 on every tetrahedron (branch-chosen logs of the shape parameters). The audit multiplies these logarithms by the full 5×12 PGL (projective general linear) gluing matrix, whose rows encode the four edge relations plus the Dehn-filling relation labelled `filling_0_0`. All four edge rows and the filling row evaluate to zero as complex intervals; the largest row diameter is 2.24×10^{-207} at α and 6.28×10^{-206} at β. Because all peripheral eigenvalues are positive, the filling relations carry no hidden $2\pi i$ term:

$$-\log M_\alpha + 2\log L_\alpha = 0, \qquad -\log M_\beta + \log L_\beta = 0.$$

Thus the closed strong flattenings (flattenings whose log-parameters sum to the value Neumann requires for the closed manifold, a strengthening of the plain flattening of Section 4) are the endpoint values of the continuous exterior Ptolemy regulator of Section 3.

Evaluating the extended Rogers sum. The extended Rogers sum is evaluated with 1000-bit real balls, together with Neumann's 6-torsion correction for a triangulation without a vertex ordering [4, Section 13 and Proposition 13.1]. The lifted flattenings satisfy Yoon's edge and Dehn-filling conditions; his Theorem 3.1 and the parity argument in the proof of Theorem 3.4 identify the sum with the filled complex volume modulo $\pi^2/2$ [5]. The correction coefficient is exactly -6 at both endpoints, contributing $-6(\pi^2/6) = -\pi^2$, which is 0 modulo $\pi^2/2$ and cancels in the difference. In the evaluator's $i(\mathrm{Vol} + i\,\mathrm{CS})$ convention, where Vol and CS denote the hyperbolic volume and Chern–Simons invariant of the filled manifold, the sums are

$$\frac{S_\alpha}{\pi^2} = -1.7333333333333333333333333\ldots, \tag{25}$$

$$\frac{S_\beta}{\pi^2} = -1.6862745098039215686627451\ldots. \tag{26}$$

The lattice selects a unique value. The Arb enclosure of $(S_\beta - S_\alpha)/\pi^2$ contains $4/85$ and has diameter less than 5×10^{-300}, far below the lattice spacing 5.61×10^{-8} of (24). Both neighboring lattice points $4/85 \pm 1/(2N)$ are rigorously excluded. Hence the enclosure contains exactly one admissible regulator value, so

$$S_\beta - S_\alpha = \frac{4\pi^2}{85} \qquad (\mathrm{mod}\ \pi^2/2). \tag{27}$$

The same lattice argument selects the individual representatives

$$S_\alpha = -\frac{26\pi^2}{15}, \qquad S_\beta = -\frac{86\pi^2}{51} \qquad (\mathrm{mod}\ \pi^2/2)$$

(consistent with (25)–(26), since $-26/15 = -1.7333\ldots$ and $-86/51 = -1.686274\ldots$).

An independent numerical cross-check. The original certificate also runs a separate numerical path that does not feed the lattice proof. It expands the actual signed Ptolemy coordinates, uses SnapPy's public cross-ratio and flattening code, and multiplies the logarithmic gluing matrix explicitly. This path uses the Vol $+i$ CS convention (rather than $i(\mathrm{Vol}+i\,\mathrm{CS})$) and returns

$$\frac{\mathrm{CS}_\alpha}{\pi^2} = \frac{1}{15}, \qquad \frac{\mathrm{CS}_\beta}{\pi^2} = \frac{1}{51} \qquad (\mathrm{mod}\ 1/6),$$

so $\mathrm{CS}_\alpha - \mathrm{CS}_\beta = 4\pi^2/85$ to about 180 decimal places (with Vol and CS as above, the hyperbolic volume and Chern–Simons invariant).

6 From the regulator to the challenge integral

Two things remain: to check that the regulator difference is genuinely the integral of η (not some other 1-form), and to remove the modulo $\pi^2/2$ ambiguity. The first is a differential identity built on the Dehn invariant and the extended Rogers dilogarithm; the second is a coarse but rigorous size bound.

The extended Rogers dilogarithm and the Dehn invariant. Let $S(M, L)$ be the continuous extended Rogers sum on the Ptolemy lift (17), with logarithms continued from the left endpoint; its endpoint values are the S_α, S_β of Section 5. The unordered-triangulation correction is constant, so its differential is zero. Neumann's *extended Rogers function* $R(z)$ (on the extended Bloch group, unrelated to the polynomial $R(s, u)$ of Section 2) has differential

$$2\,dR(z) = \log z\, d\log(1-z) - \log(1-z)\, d\log z. \tag{28}$$

The remaining ingredient is the Dehn invariant, which measures how the shapes vary along the peripheral torus. For $n = 2$, Zickert's Dehn-invariant formula gives $-2\,M \wedge L$ in a peripheral basis of intersection number $+1$ [6, Theorem 1.10], where $M \wedge L$ is the wedge of the meridian and longitude eigenvalues in the target of the Dehn invariant. Our M, L are the challenge and SnapPy standard-basis eigenvalues by the exact identity $L\lambda_{11} = 1$ from Section 2, and the standard knot meridian–longitude basis has intersection number -1 [6, Section 3.1]. In that basis the Dehn invariant is therefore

$$\nu = 2\,M \wedge L. \tag{29}$$

As a direct sign check, the independent script `dehn_wedge_reconstruction.py` forms $z_i \wedge (1 - z_i)$ from the exponent vectors of all 24 signed-coordinate monomials; every wedge involving the Ptolemy variables b, c, e cancels, leaving the single surviving coefficient $+2$ on $M \wedge L$.

Remark 3. The sign in (29) is no longer accepted only from the original certificate. The independent reconstruction enumerates the orientation signs, cancels every wedge involving the internal Ptolemy variables, and leaves the single coefficient $+2$ on $M \wedge L$.

The differential is exactly η. Applying (28) to the formal shape sum and adding the Dehn term (29) gives

$$dS = \log M\, d\log L - \log L\, d\log M = \eta. \tag{30}$$

Since the continuous Ptolemy lift connects the exact endpoint flattenings, integrating (30) and using (27) yields the integral modulo the period,

$$I := \int_\alpha^\beta \eta = \frac{4\pi^2}{85} \qquad (\mathrm{mod}\ \pi^2/2). \tag{31}$$

(This integral I is unrelated to the identity matrix I of Section 2.)

Removing the period ambiguity. The stated branch fixes the real lift with no numerical quadrature. Put $X = -\log x$ and $Y = -\log y$, both positive since $x, y \in (0,1)$. (This X is unrelated to the fixed-field generator X of Section 4.) Since $dx > 0$ and $dy < 0$ along the branch,

$$\eta = X\,dY - Y\,dX > 0,$$

so $I > 0$. Exact root isolation gives

$$\frac{1}{4} < \alpha < \frac{3}{8} < \beta < \frac{1}{2}, \qquad \frac{1}{4} < y(\beta) < \frac{1}{2} < y(\alpha) < 1,$$

with endpoint values $y(\alpha) = \sqrt{\alpha} \approx 0.5909894287$ and $y(\beta) = \beta \approx 0.4068130813$. It follows that

$$\begin{aligned} I &< \log 4\big((Y_\beta - Y_\alpha) + (X_\alpha - X_\beta)\big) \\ &= \log 4\,\log\!\left(\frac{y(\alpha)}{y(\beta)}\frac{\beta}{\alpha}\right) < \log 4\,\log 8 < \frac{9}{2} < \frac{\pi^2}{2}. \end{aligned}$$

Both I and $4\pi^2/85$ lie strictly between 0 and $\pi^2/2$, so the congruence (31) has exactly one admissible real lift. This proves the challenge identity:

$$\boxed{\int_\alpha^\beta \left(\log x\,\frac{dy}{y} - \log y\,\frac{dx}{x}\right) = \frac{4\pi^2}{85}.}$$

7 Reproducibility

The public release uses the independent-verification route rather than requiring SnapPy's private `dev.extended_ptolemy` interface. It requires Python with `sympy` and `mpmath`, plus SageMath 10.9 with SnapPy 3.3.2 installed in Sage's Python. From the repository root, the complete gate is

```
verification/run_all.sh
```

The runner first checks that Sage and SnapPy share one interpreter, then executes the exact Riley identities, independent endpoint and uniqueness checks, Dehn-wedge reconstruction, direct quadrature, by-hand triangulation and flattening reconstruction, and the final Arb lattice-selection certificate. The load-bearing final command is

```
$SAGE -python verification/exactness_closure.py
```

and prints its explicit closed verdict only after the field, torsion-bound, interval, and neighboring-lattice-point assertions pass. The exact status of every layer is recorded in the verification ledger shipped with the source. Borel's finiteness theorem and the Zickert–Neumann regulator normalizations are cited inputs, not reproved.

Acknowledgments and tooling disclosure

The Ramanujan Machine group posed the problem. The project was developed by Robert Sneiderman with extensive assistance from OpenAI Codex and other frontier language models for symbolic exploration, implementation, adversarial review, and exposition. Model output is not treated as mathematical evidence: the claim rests on the displayed argument, cited published theorems, and the reproducible exact and certified computations described above.

References

[1] D. Cooper, M. Culler, H. Gillet, D. D. Long, P. B. Shalen, *Plane curves associated to character varieties of* 3*-manifolds*, Invent. Math. **118** (1994), no. 1, 47–84.

[2] A. Borel, *Cohomologie de* SL_n *et valeurs de fonctions zêta aux points entiers*, Ann. Scuola Norm. Sup. Pisa Cl. Sci. (4) **4** (1977), no. 4, 613–636.

[3] V. T. Khoi, *On the integral of* $\log x\, dy/y - \log y\, dx/x$ *over the A-polynomial curves*, Acta Math. Vietnam. **33** (2008), no. 3, 519–528; arXiv:0811.2725.

[4] W. D. Neumann, *Extended Bloch group and the Cheeger–Chern–Simons class*, Geom. Topol. **8** (2004), 413–474; arXiv:math/0307092.

[5] S. Yoon, *The volume and Chern–Simons invariant of a Dehn-filled manifold*, Topology Appl. **256** (2019), 105–121; arXiv:1801.08288.

[6] C. K. Zickert, *Ptolemy coordinates, Dehn invariant and the A-polynomial*, Math. Z. **283** (2016), 515–537; arXiv:1405.0025.

[7] C. K. Zickert, *The extended Bloch group and algebraic K-theory*, J. Reine Angew. Math. **704** (2015), 21–54; arXiv:0910.4005.